Universidad Internacional del Ecuador (UIDE)
Computer Science School
Network Communication Technology
Title: *Phase 1 – Network Infrastructure Planning Report: Enterprise Chat Integrated Project (GTA1_Phase_1_Integrated Project)*
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Repository: https://github.com/ales231/Integrative_project_
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1. Introduction

1.1 Context

The Integrated Project combines network engineering, a Go back end, a FastAPI service, statistics and technology-and-society analysis over sixteen weeks. Our team chose the **enterprise chat** option as the application: a Go server that handles real-time messaging and a FastAPI service that authenticates users and issues tokens. Both must run on a network with two sites: Site A, the headquarters (HQ) in Quito, and Site B, a branch office in Guayaquil, designed from the reference architecture of the project statement. This document is the Phase 1 deliverable: it plans the network infrastructure and presents a preliminary design. As the assignment states, a fully configured or operational network is not required at this phase.

1.2 Objectives

- Describe the two sites, their users, estimated devices, services and communication needs.
- Define how OpenWrt, VLANs, VLSM, 802.1Q trunks, Router-on-a-Stick, EtherChannel, RSTP, OSPF, DHCP, NAT and the firewall will be used to meet those needs.
- Present physical and logical topologies and distribute the 10.10.0.0/16 block with VLSM.
- Locate the Go application and the FastAPI authentication service and list the flows they require.
- Set a work plan (activities, schedule, owners, risks) and a validation plan with planned evidence.

1.3 Assumptions

Where the project statement leaves a value open, the team made the following assumptions. They are proposals and can be adjusted without changing the overall design.

- Two sites: Site A (HQ, Quito) and Site B (branch, Guayaquil). Each has one OpenWrt router on a Raspberry Pi (or an equivalent platform authorized by the instructor). Both routers join the shared Wi-Fi segment 192.168.100.0/24 (access point .1, OpenWrt-A .10, OpenWrt-B .11) and each site reaches the Internet independently through that access point, as in the reference architecture.
- Each site has two Cisco switches (SW-A1/SW-A2 and SW-B1/SW-B2) with an IOS command line. The VLAN IDs follow the reference: 10 Users, 20 Servers, 30 Guests and 99 Administration at both sites.
- Device counts are estimates. Each subnet is sized for the current count plus 25 % growth, plus one address for the gateway.
- Chat traffic is light (short text messages over persistent connections); the shared Wi-Fi segment is assumed to provide at least 10 Mbit/s between the sites.
- Switch names and port numbers are planning labels and may change during implementation.

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2. Proposed Methodology

The team follows four steps: (1) analyze the scenario and requirements, (2) design the addressing plan and topology, (3) plan the application integration and the work, and (4) define how every requirement will be validated. Sections 2.1 to 2.6 describe these planning decisions; Section 3 presents the resulting topology and addressing tables.

2.1 Scenario and scope

The fictitious company has about 100 employees at Site A (HQ) and 35 at Site B (branch). We estimate **276 devices at Site A** and **96 at Site B** today (laptops, desktops, IP phones, printers, access points, servers and network equipment), growing to 345 and 120 with the 25 % allowance. Table 1 summarizes each site.

Table 1

Sites in Scope

Site	Users / role	Estimated devices (now → planned)	Main services
Site A – HQ, Quito	Sales and support, administration and finance, development and IT; data-centre role for the chat platform	276 → 345	Go chat server, FastAPI authentication, PostgreSQL, DHCP, DNS, NAT, firewall, guest Wi-Fi, own Internet access
Site B – Branch, Guayaquil	Staff and sales/support; local services	96 → 120	DHCP, DNS, NAT, firewall, own Internet access; chat and login through Site A

Note. Device counts are planning estimates; “planned” adds a 25 % growth allowance.

Communication needs.

- **Chat:** every employee client connects to the Go chat server at Site A over HTTPS/WebSocket and authenticates with the FastAPI service. Site B users cross the shared Wi-Fi segment for both.
- **Internet:** each site reaches the Internet on its own through the shared access point; employees need web access and software updates, and guests get Internet only.
- **Segmentation:** servers, administration, users and guests must be isolated and reachable only as the firewall policy allows.
- **Infrastructure:** automatic addressing (DHCP), name resolution (DNS), time (NTP) and administration by SSH/HTTPS from the administration VLAN only.
- **Resilience:** the Layer-2 core of each site must survive the loss of one EtherChannel member and must not form loops.

Scope. In scope for this phase: requirements, VLAN and addressing plan, physical and logical topology, service and security design, application placement, work plan and validation plan. Out of scope: full device configuration, application implementation and the Wi-Fi radio design (the access point is provided by the lab). As the project statement says, high availability, load balancing, cloud, SDN, IPv6 and routing protocols

other than OSPF are not required.

2.2 Network requirements

Table 2 states, for each required technology, the role it plays here and the design decision. The sources behind them are VLAN trunking and EtherChannel (Cisco, 2007), RSTP (Cisco, 2023), OSPFv2 (FRRouting Project, n.d.; Moy, 1998), DHCP (Droms, 1997) and NAT (Srisuresh & Egevang, 2001).

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Table 2

Technologies and Design Decisions

Technology	Role in this network	Design decision
OpenWrt	Router, DHCP/DNS, firewall, NAT and OSPF platform at both sites (OpenWrt Project, n.d.-a, n.d.-b).	OpenWrt-A and OpenWrt-B on Raspberry Pi. Configuration in /etc/config/network, dhcp and firewall; OSPF through the FRRouting packages (frr, frr-zebra, frr-ospfd) (FRRouting Project, n.d.). wlan0 (client of the shared access point) is the WAN and inter-site interface; eth0 is the LAN trunk. Administration by SSH only from VLAN 99.
VLANs	Separate broadcast domains by function and trust level.	4 VLANs per site with the IDs of the reference architecture: 10 Users, 20 Servers, 30 Guests, 99 Administration. The IDs repeat at both sites but Layer 2 is not extended between them. Native VLAN 999 (unused, no IP) on trunks; unused ports shut down (OpenWrt Project, n.d.-c).
VLSM	Fit each subnet to its device count without waste.	Largest subnet first inside each site block (Section 2.3, Tables 11 and 12): /24, /25, /27 and /27 at Site A; /25, /27, /28 and /28 at Site B. Each site can be summarized as one /22 (Fuller & Li, 2006).
802.1Q trunks	Carry several VLANs over one link.	Trunks: router eth0 – SW-A1 (SW-B1) and between the two switches of each site. Allowed VLANs limited to 10, 20, 30 and 99; dynamic trunk negotiation disabled (Cisco, 2007).
Router-on-a-Stick	Inter-VLAN routing with one physical router port.	One trunk port (eth0) on each OpenWrt router with one 802.1Q sub-interface per VLAN (eth0.10, eth0.20, eth0.30, eth0.99; OpenWrt Project, n.d.-c). The gateway is the first usable address of each subnet. Inter-VLAN traffic crosses the trunk twice, so the 1 Gbit/s trunk is a known bottleneck and single point of failure (see risks).
EtherChannel	Add bandwidth and resilience between switches.	Po1 between the two switches of each site: four 1 Gbit/s links, LACP (IEEE 802.3ad) in active mode on both ends (Cisco, 2007). As in the reference, an optional backup bundle Po2 with PAgP (Cisco) uses four other ports, because LACP and PAgP cannot form one bundle; RSTP keeps Po2 blocked while Po1 is up.
RSTP	Loop prevention with fast convergence.	Rapid Spanning Tree on all switches. SW-A1 and SW-B1 are root bridges (priority 4096); SW-A2 and SW-B2 are backup roots (8192). Access ports use PortFast and BPDU Guard (Cisco, 2023).
OSPF	Dynamic routing between sites.	OSPFv2, single area 0, over the shared Wi-Fi segment 192.168.100.0/24 (wlan0 on both routers) with loopback router IDs (10.10.255.1 and .2). The VLAN subnets of each site are announced and the VLAN sub-interfaces are passive. The default route is not advertised: each site keeps its own default route to the access point (FRRouting Project, n.d.; Moy, 1998). Authentication on the segment is planned as a hardening step.
DHCP	Automatic addressing for user VLANs.	dnsmasq on each OpenWrt router, one scope for the Users VLAN and one for the Guests VLAN (Table 13). Servers and Administration use static addresses. DNS names for services (chat, auth, db) are served by the same dnsmasq (OpenWrt Project, n.d.-a).
NAT	Let private 10.10.0.0/16 hosts reach the Internet.	Source NAT (masquerade) on the wan zone (wlan0) of each router, so every site reaches the Internet independently through the shared access point (Srisuresh & Egevang, 2001; Rekhter et al., 1996). Masquerading excludes destinations in 10.10.0.0/16 (a negated masq_dest), so inter-site traffic keeps its real source address (OpenWrt Project, n.d.-b). No port forwarding is planned in Phase 1.
Firewall	Enforce segmentation between zones.	OpenWrt zones (OpenWrt Project, n.d.-b): mgmt (VLAN 99), srv (20), users (10), guest (30) and wan (wlan0, which carries both Internet and inter-site traffic). Default policy reject for forwarding; explicit allow rules (Table 3). Servers also run a host firewall, because traffic inside one VLAN does not cross the router.

Note. Switch names and port numbers are planning labels.

Table 3

Planned Firewall Policy Between Zones

#	Source zone / subnet	Destination	Service	Action
1	users (VLAN 10)	srv: chat 10.10.1.130, auth 10.10.1.131	tcp/443, tcp/8000	Allow
2	users, srv (restricted)	wan: Internet	tcp/80, tcp/443, DNS, NTP	Allow + NAT
3	mgmt (VLAN 99)	all local zones	SSH, HTTPS, ICMP	Allow
4	users	mgmt	any	Deny
5	guest (VLAN 30)	wan: Internet only	tcp/80, tcp/443, DNS	Allow + NAT
6	guest	any internal zone and 10.10.0.0/16	any (except DNS/DHCP to router)	Deny
7	wan: Site B users 10.10.4.0/25	srv: chat and auth at Site A	tcp/443, tcp/8000	Allow (same as rule 1)
8	wan: peer router (192.168.100.11 / .10)	router	OSPF (protocol 89), ICMP	Allow
9	wan: 10.10.0.0/16 (other site)	users, srv, mgmt	ICMP echo (tests)	Allow
10	wan: any other source	any	any (except replies to outbound traffic)	Drop

Note. The default policy is reject for forwarding; replies to allowed connections are accepted automatically. Rules are written for Site A; Site B applies the same rules, and its rule 2 also lets users reach chat and auth at Site A (10.10.1.130–131) through wan without NAT.

2.3 Addressing strategy

- Private space 10.10.0.0/16 (Rekhter et al., 1996), divided hierarchically: one /22 per site, so each site can be advertised or summarized as a single prefix.
- VLSM inside each site: subnets sorted from largest to smallest and allocated contiguously on their natural boundaries, so no subnets overlap (Fuller & Li, 2006).
- Sizing rule: hosts required = devices × 1.25 + 1 (gateway); choose the smallest prefix with $2^n - 2 \geq$ hosts required.
- Gateway = first usable address; the next nine host addresses are reserved for static infrastructure; DHCP pools start at host .11 of the subnet.

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Table 4

Distribution of the Address Space

Block	Purpose	Notes
10.10.0.0/22	Site A – HQ, Quito	4 VLANs; allocated to 10.10.1.191; free from 10.10.1.192 to 10.10.3.255 for growth
10.10.4.0/22	Site B – Branch, Guayaquil	4 VLANs; allocated to 10.10.4.191; free from 10.10.4.192 to 10.10.7.255 for growth
10.10.8.0 – 10.10.254.255	Reserved	Future sites, DMZ or new VLANs (not allocated)
10.10.255.0/24	Loopbacks / router IDs	10.10.255.1/32 OpenWrt-A; 10.10.255.2/32 OpenWrt-B
192.168.100.0/24	Shared Wi-Fi segment (outside 10.10.0.0/16)	Provided by the access point: AP .1, OpenWrt-A .10, OpenWrt-B .11; OSPF area 0

The per-VLAN subnets (Tables 11 and 12) and the DHCP scopes (Table 13) are in Section 3.2.

2.4 Integration plan: Go application and FastAPI authentication

Both services are placed at Site A in the servers VLAN 20 (10.10.1.128/27), behind the router firewall, so that the single copy of each service is reachable from the Users VLAN of Site A and, over OSPF routes learned on the shared Wi-Fi segment, from Site B. No Internet exposure is planned in Phase 1.

Table 5

Application Services and Placement

Service	Address / name	Port	Purpose
Go chat server	10.10.1.130 – chat.corp.local	tcp/443 (HTTPS, WebSocket upgrade)	Real-time messaging over persistent WebSocket connections (Fette & Melnikov, 2011) (The Go Authors, n.d.)
FastAPI authentication	10.10.1.131 – auth.corp.local	tcp/8000	Login, token issue and token validation endpoint (Ramírez, n.d.); JSON Web Tokens (Jones et al., 2015)
PostgreSQL	10.10.1.132 – db.corp.local	tcp/5432	Users and message storage; reachable only from the two servers

Note. All three services are in VLAN 20 at Site A.

Table 6

Communication Flows Required by the Application

Flow	Source → destination	Port	Allowed by
Login (obtain token)	Client (Users VLAN, either site) → FastAPI	tcp/8000	Firewall rules 1 and 7
Chat session	Client → Go chat server (HTTPS, then WebSocket)	tcp/443	Firewall rules 1 and 7
Token validation	Go chat server → FastAPI (same VLAN)	tcp/8000	Host firewall on servers
Persistence	Go chat and FastAPI → PostgreSQL (same VLAN)	tcp/5432	Host firewall on servers
Name resolution	Clients → OpenWrt dnsmasq of their site	udp/tcp 53	Router input rule per zone
Administration	Administration VLAN 99 → servers	tcp/22	Firewall rule 3
Updates	Servers → Internet via NAT at Site A	tcp/80, tcp/443	Firewall rule 2

Note. Firewall rule numbers refer to Table 3.

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Requirements the network must meet.

- **Reachability:** routes to 10.10.1.128/27 must be present at Site B (learned by OSPF over wlan0) and the return path must not be NAT-ed.
- **Long-lived connections:** firewall and connection-tracking idle timeouts must be longer than the WebSocket ping interval (planned 30 s), so chat sessions are not cut silently.
- **Latency:** the chat should stay interactive across the shared Wi-Fi segment; round-trip time and packet loss between sites will be measured in the validation plan.
- **Time and DNS:** servers and routers synchronize time by NTP so token expiry is consistent; clients use names, not IP addresses, so servers can be moved later.
- **Transport security:** TLS on tcp/443 and on the authentication endpoint (an internal certificate authority is acceptable); PostgreSQL is not published outside VLAN 20.
- **Single point of failure:** if the shared Wi-Fi segment or OpenWrt-A fails, Site B users lose chat (Site B keeps its own Internet access). A replica at Site B is listed as future work.

Authentication sequence. (1) The client sends credentials to the FastAPI service; (2) FastAPI verifies them against PostgreSQL and returns a signed JSON Web Token; (3) the client opens a WebSocket to the Go server presenting the token; (4) the Go server validates the token with FastAPI (or by verifying its signature) and accepts or rejects the session. Only steps 1 and 3 cross VLAN or site boundaries.

2.5 Work plan

The plan follows the sixteen-week project. Weeks are relative to the project start; work begins in Week 2 and calendar dates will be aligned with the course schedule. S1, S2 and S3 are the three team members, numbered as on the cover (roles below).

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Figure 1

Project Schedule by Activity and Owner

Activity	Owner	Week															
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A1 Scenario and requirements	S1		■	■													
A2 VLSM addressing plan	S1			■	■												
A3 Topology design	S1				■	■											
A4 Security, DHCP, NAT policy	S2				■	■											
A5 Integration design, repository	S3				■	■	■										
A6 Phase 1 report	All						■										
A7 Layer 2 build	S1							■	■								
A8 Layer 3 build	S1, S2								■	■	■						
A9 NAT and firewall	S2										■	■					
A10 Deploy Go and FastAPI	S3										■	■	■				
A11 Validation and evidence	All												■	■	■		
A12 Final documentation, demo	All															■	■

Note. Shaded cells mark the project weeks in which each activity takes place; Week 1 is left empty because work begins in Week 2. S1–S3 are the three team members and "All" is the whole team.

Table 7

Activities, Dependencies, and Resources

ID	Activity	Weeks	Owner	Depends on	Resources	Deliverable
A1	Scenario, requirements and assumptions	2–3	S1	Project statement	Project document	Sections 1–2.2
A2	VLSM addressing plan	3–4	S1	A1	Manual calculation	Tables 4, 11–13
A3	Physical and logical topology	4–5	S1	A1, A2	Diagram tools	Figures 2–3
A4	Security, DHCP and NAT policy	4–5	S2	A1, A2	OpenWrt docs	Table 3, scopes
A5	Integration design and repository	4–6	S3	A1	Git repository	Section 2.4, repo structure
A6	Phase 1 report and team review	6	All	A1–A5	—	This PDF
A7	Layer 2: VLANs, trunks, LACP, RSTP	7–8	S1	A3, A6	Cisco switches (lab)	Switch configs
A8	Layer 3: OpenWrt ROAS, DHCP, OSPF	8–10	S1, S2	A7	Raspberry Pi, OpenWrt, FRR	Router configs
A9	NAT and firewall zones	10–11	S2	A8	OpenWrt fw4	Rule set
A10	Deploy Go chat and FastAPI	10–12	S3	A8	Linux VMs/containers	Running services
A11	Validation tests and evidence	12–14	All	A9, A10	Wireshark, CLI tools	Evidence folder
A12	Final documentation and demo	15–16	All	A11	—	Final report, demo

Note. S1–S3 are the three team members; weeks are relative to the project start. ROAS = Router-on-a-Stick.

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Responsibilities.

- **S1 – Luis Angel Novillo Cruz, network design lead:** addressing, VLANs, trunks, EtherChannel, RSTP, OSPF, topology diagrams.
- **S2 – Arthur Brayan Beltran Hernández, security and services lead:** OpenWrt firewall zones, NAT, DHCP/DNS, hardening.
- **S3 – Alex Fernando Alban Torres, integration and validation lead:** Go and FastAPI deployment, repository, test scripts and evidence.
- All members review each deliverable and keep configurations and evidence in the repository.

Technical risks.

Table 8

Risk Register

Risk	Impact	Mitigation
Router-on-a-Stick trunk is a bottleneck and a single point of failure	Inter-VLAN throughput limited; outage if the trunk fails	Keep heavy server traffic inside VLAN 20; monitor trunk load; consider a second trunk or a Layer-3 switch as future work
FRRouting is missing or too heavy on the Raspberry Pi OpenWrt image	No dynamic routing	Test FRR on the image in Week 7; fallback: static routes for the two sites
The shared Wi-Fi segment is unreliable, or the access point isolates clients and blocks multicast	OSPF neighbors do not form; high latency or loss between sites	Test OSPF over wlan0 early; tune hello and dead timers; fallback: unicast neighbors or static routes; measure RTT and loss
Wrong RSTP root, native-VLAN mismatch, or LACP and PAgP mixed on the same ports	Broadcast storm, lab outage	Set root priorities explicitly; keep LACP and PAgP bundles on separate ports; BPDU Guard on edge ports; verify with spanning-tree output before adding VLANs
NAT applied to inter-site traffic	Real source addresses hidden; firewall rules fail	Exclude 10.10.0.0/16 from masquerading; verify with test T8
Firewall rule locks administrators out	Loss of device access	Keep console access; apply mgmt rules first; back up configs in the repository
Idle timeouts drop WebSocket sessions	Chat disconnects	Application ping every 30 s; compare against timeouts; test with a long-running session
Limited lab access to Raspberry Pi and Cisco switches	Delays or an unstable lab	Book lab time early; prepare and back up configurations in the repository; keep a minimal topology profile
Clock skew between servers	Tokens rejected as expired or not yet valid	NTP on all servers and routers
Team availability and late dependencies	Schedule slip	Weekly check-in; dependencies in Table 7; buffer in Weeks 15–16

2.6 Validation plan

Each requirement is checked by at least one test with a defined expected result and a saved piece of evidence. Commands are indicative; names will follow the final devices.

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Table 9

Planned Validation Tests

ID	Requirement	Test	Expected result	Evidence
T1	VLSM	Script checks that subnets do not overlap and fit their blocks; ping each gateway	No overlap; all 8 gateways answer	Script output, ping log
T2	VLANs, 802.1Q trunks	show vlan brief; show interfaces trunk on every switch	VLANs 10, 20, 30, 99 present; trunks list only allowed VLANs; native 999	CLI captures
T3	Router-on-a-Stick	Ping and traceroute between hosts of different VLANs at one site	Reply; one router hop; traffic seen tagged on the trunk	Traceroute, Wireshark capture
T4	EtherChannel	show etherchannel summary; unplug one of the four Po1 members while pinging	Po1 up with four ports bundled; at most one or two packets lost	CLI capture, ping log
T5	RSTP	show spanning-tree on the switches of each site	Root is SW-A1 (SW-B1); no loops; the optional Po2, if built, stays blocked and forwards when Po1 fails	CLI captures, ping log
T6	OSPF	show ip ospf neighbor; show ip route ospf on both routers; drop wlan0 and restore	Neighbor Full; each site learns the other /22; no default route learned by OSPF; routes recover	vtysh output
T7	DHCP	Connect a client per Users VLAN and Guests VLAN at each site; read lease	Address inside scope, correct gateway and DNS, lease time as planned	ip addr / lease file
T8	NAT	Capture on wlan0 while a Site A host browses and while a Site B host opens the chat	Internet traffic leaves with the router wlan0 address; traffic to 10.10.0.0/16 keeps its real source	Capture on wlan0
T9	Firewall	Test allowed and denied flows from Table 3 (ping, nc, curl, nmap)	Allowed flows pass; guest to internal blocked; user to mgmt blocked; wan inbound dropped	Test matrix, rule listing
T10	Addressing for services	Resolve chat, auth and db names from both sites	Names resolve to 10.10.1.130–132	nslookup output
T11	Application reachability	From each site, request a token from FastAPI and open a WebSocket to the Go server	Token issued; WebSocket accepted; Site B behaves like Site A	curl output, client log
T12	Latency and idle timeout	Round-trip time and loss between sites; keep a WebSocket idle for 15 minutes	RTT recorded; session survives the idle period	Ping stats, session log
T13	Segmentation of PostgreSQL	Try tcp/5432 from the Users VLAN and from the servers VLAN	Refused from users; accepted only from the two servers	nc/psql output

Note. Commands are indicative; device names follow the final topology.

Evidence handling.

- Evidence is stored in the repository under an `evidence/` folder, one subfolder per test ID (T1–T13), with dated screenshots or text captures, pcap files and short notes.
- Final configurations of every device are committed; each change is traceable in the Git history.
- A results table (test ID, date, tester, pass/fail, link to evidence) is kept in the repository and attached to the final report.

- Failed tests are logged with cause and fix, then repeated; the whole set is rerun before delivery.

3. Results / Development

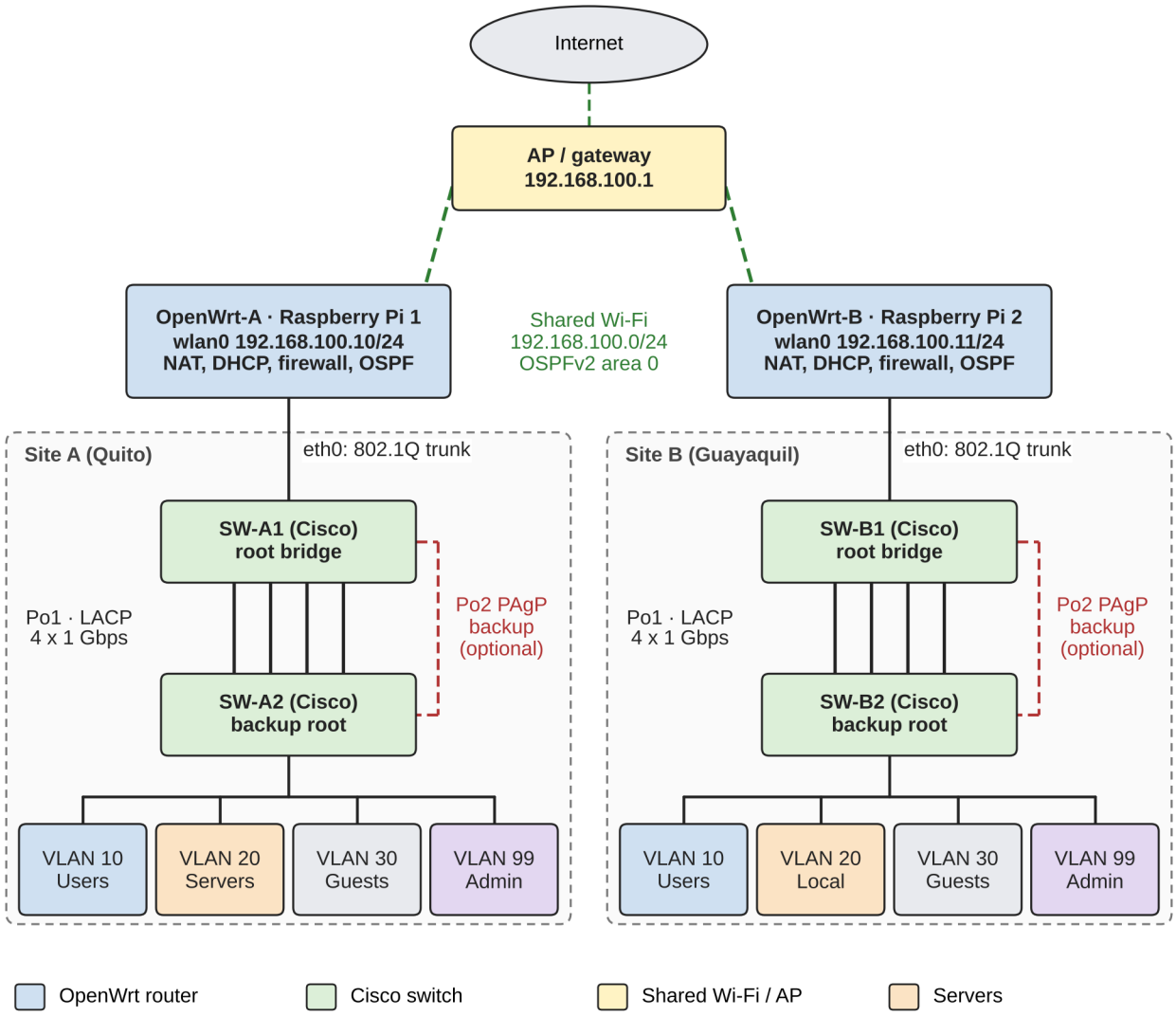
3.1 Preliminary topology

Figure 2 shows the physical design, adapted from the reference architecture. At each site an OpenWrt router on a Raspberry Pi joins the shared Wi-Fi segment through wlan0 and reaches its switches through eth0, a single 802.1Q trunk (Router-on-a-Stick). Two Cisco switches per site are bonded with a four-link LACP EtherChannel; the first switch (SW-A1, SW-B1) is the root bridge and the second (SW-A2, SW-B2) carries the access ports of the four VLANs. An optional PAgP bundle serves as a backup. The two routers exchange routes by OSPF over the Wi-Fi segment, and each site reaches the Internet on its own through the same access point.

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Figure 2

Physical Topology of Both Sites



Note. The four parallel lines are the Po1 EtherChannel (LACP, 4 × 1 Gbit/s); the dashed red line is the optional PAgP backup bundle, blocked by RSTP; dashed green lines are Wi-Fi. Device and port names are planning labels.

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Table 10

Planned Links

Link	Endpoints	Type	Notes
L1	Access point – Internet	Uplink	Provided by the lab; shared by both sites
L2	OpenWrt-A wlan0 – access point	Wi-Fi client, 192.168.100.10/24	WAN, NAT and OSPF area 0 (Site A)
L3	OpenWrt-A eth0 – SW-A1	802.1Q trunk	VLANs 10, 20, 30, 99; native 999
L4	SW-A1 – SW-A2 (4 links)	EtherChannel Po1 (LACP)	Trunk, same allowed VLANs
L5	SW-A1 – SW-A2 (4 links, optional)	EtherChannel Po2 (PAgP)	Backup; blocked by RSTP while Po1 is up
L6	SW-A2 – servers	Access, VLAN 20	Go chat, FastAPI, PostgreSQL
L7	OpenWrt-B wlan0 – access point	Wi-Fi client, 192.168.100.11/24	WAN, NAT and OSPF area 0 (Site B)
L8	OpenWrt-B eth0 – SW-B1	802.1Q trunk	VLANs 10, 20, 30, 99; native 999
L9	SW-B1 – SW-B2 (4 links)	EtherChannel Po1 (LACP)	Trunk, same allowed VLANs
L10	SW-B1 – SW-B2 (4 links, optional)	EtherChannel Po2 (PAgP)	Backup; blocked by RSTP while Po1 is up

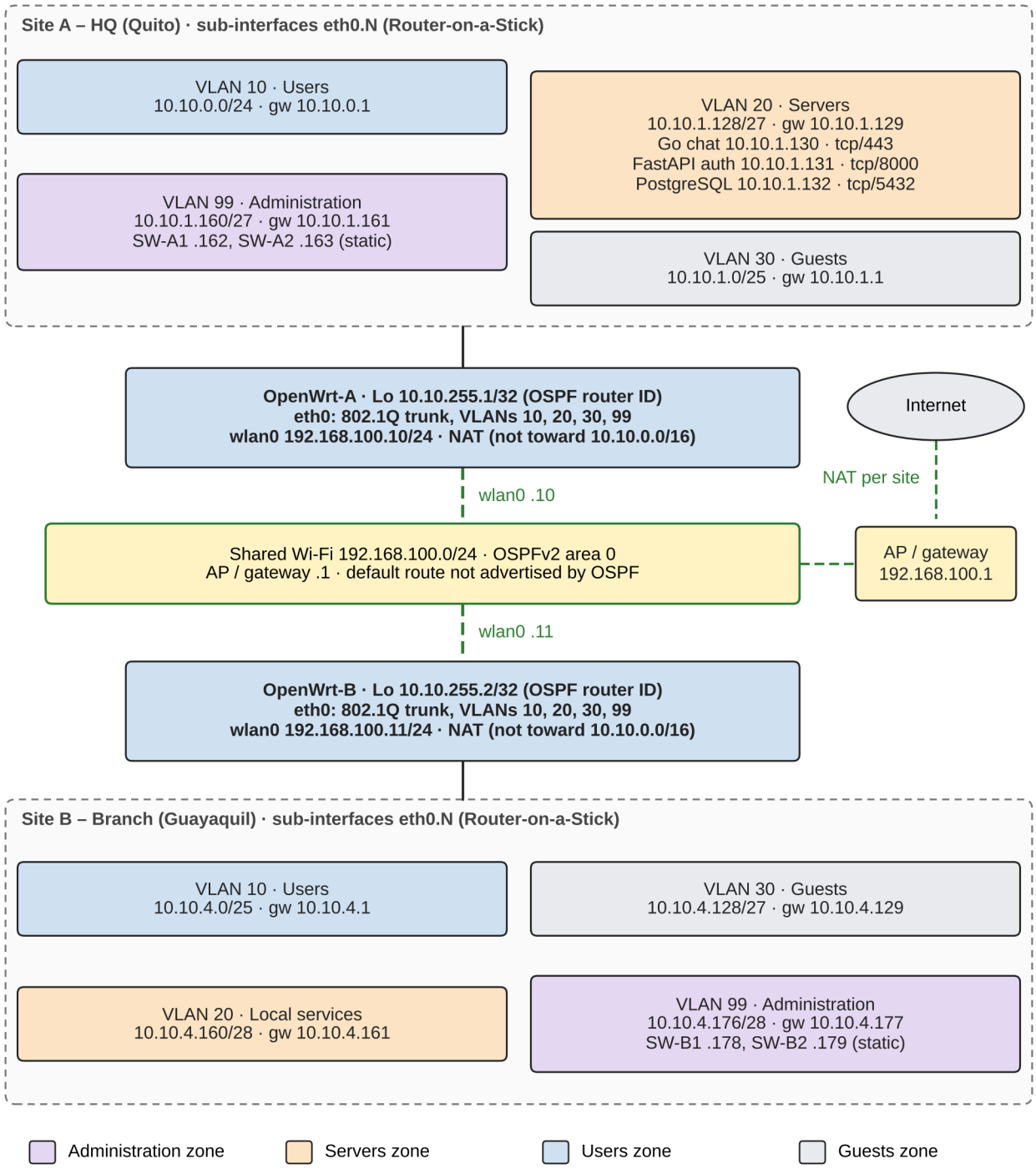
Note. Port and device names are planning labels. LACP = Link Aggregation Control Protocol.

Figure 3 shows the Layer-3 view: each router holds one sub-interface per VLAN (eth0.10, eth0.20, eth0.30, eth0.99), a loopback used as OSPF router ID, and wlan0 on the shared Wi-Fi segment 192.168.100.0/24. Colors indicate the firewall zone of each VLAN. The application services sit in the servers VLAN of Site A.

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Figure 3

Logical (Layer 3) Topology, VLANs, Subnets, and Application Placement



Note. Colors show firewall zones: purple, administration; orange, servers; blue, users; gray, guests. Subnets and gateways match Tables 11 and 12.

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3.2 Addressing results (VLSM)

Tables 11 and 12 list every subnet in allocation order. The values were calculated by hand with the sizing rule of Section 2.3 and cross-checked with a script (Python `ipaddress` module): no overlap, each subnet inside its site block, and enough usable hosts for the planned count.

Table 11

VLSM Plan for Site A (HQ, Quito), Block 10.10.0.0/22

VLAN	Name	Now	Plan	CIDR	Mask	Gateway	Usable host range	Broadcast	Use
10	Users	164	205	10.10.0.0/24	255.255.255.0	10.10.0.1	10.10.0.1 – 10.10.0.254	10.10.0.255	81%
30	Guests	80	100	10.10.1.0/25	255.255.255.128	10.10.1.1	10.10.1.1 – 10.10.1.126	10.10.1.127	80%
20	Servers (Go, FastAPI, DB)	16	20	10.10.1.128/27	255.255.255.224	10.10.1.129	10.10.1.129 – 10.10.1.158	10.10.1.159	70%
99	Administration	16	20	10.10.1.160/27	255.255.255.224	10.10.1.161	10.10.1.161 – 10.10.1.190	10.10.1.191	70%

Note. Now = estimated current devices; Plan = devices plus 25 % growth; Use = (planned devices + gateway) ÷ usable hosts.

Table 12

VLSM Plan for Site B (Branch, Guayaquil), Block 10.10.4.0/22

VLAN	Name	Now	Plan	CIDR	Mask	Gateway	Usable host range	Broadcast	Use
10	Users	60	75	10.10.4.0/25	255.255.255.128	10.10.4.1	10.10.4.1 – 10.10.4.126	10.10.4.127	60%
30	Guests	20	25	10.10.4.128/27	255.255.255.224	10.10.4.129	10.10.4.129 – 10.10.4.158	10.10.4.159	87%
20	Local services	8	10	10.10.4.160/28	255.255.255.240	10.10.4.161	10.10.4.161 – 10.10.4.174	10.10.4.175	79%
99	Administration	8	10	10.10.4.176/28	255.255.255.240	10.10.4.177	10.10.4.177 – 10.10.4.190	10.10.4.191	79%

Note. Now = estimated current devices; Plan = devices plus 25 % growth; Use = (planned devices + gateway) ÷ usable hosts.

Static addresses: switches SW-A1 and SW-A2 at 10.10.1.162 and 10.10.1.163 (VLAN 99); SW-B1 and SW-B2 at 10.10.4.178 and 10.10.4.179 (VLAN 99); Go chat server 10.10.1.130, FastAPI authentication 10.10.1.131 and PostgreSQL 10.10.1.132 (VLAN 20). Table 13 lists the DHCP scopes.

Table 13

DHCP Scopes Served by dnsmasq on OpenWrt

Site	VLAN	Name	Scope (first – last)	Lease	Options
A	10	Users	10.10.0.11 – 10.10.0.254	12 h	router 10.10.0.1; DNS 10.10.0.1; domain corp.local
A	30	Guests	10.10.1.11 – 10.10.1.126	1 h	router 10.10.1.1; DNS 10.10.1.1; domain corp.local
B	10	Users	10.10.4.11 – 10.10.4.126	12 h	router 10.10.4.1; DNS 10.10.4.1; domain corp.local
B	30	Guests	10.10.4.139 – 10.10.4.158	1 h	router 10.10.4.129; DNS 10.10.4.129; domain corp.local

Note. VLANs 20 and 99 use static addresses at both sites and have no DHCP scope.

4. Conclusions and Future Works

- The two-site design follows the reference architecture: one OpenWrt router (Raspberry Pi) per site doing Router-on-a-Stick, DHCP, NAT, firewall and OSPF, and two Cisco switches per site with LACP and RSTP.
- VLSM inside two /22 site blocks serves 8 VLANs (four per site), keeps a 25 % growth allowance and leaves most of 10.10.0.0/16 free for new sites.
- The Go chat server and FastAPI authentication service are placed in a dedicated servers VLAN at Site A, with explicit flows, so firewall rules and tests can be written before the application is finished.
- The main weaknesses, a single router trunk and a single shared Wi-Fi segment between sites, are known and have mitigations and tests.

Future work: add a second trunk or a Layer-3 switch; replace the shared Wi-Fi segment with a wired or redundant inter-site path with OSPF failover; place a chat/authentication replica at Site B; add 802.1X on user ports and centralized logging and monitoring; evaluate a DMZ and reverse proxy if remote access is needed.

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6. References

Cisco. (2007, November 16). *Configuring EtherChannel and 802.1Q trunking between Catalyst L2 fixed configuration switches and Catalyst switches running CatOS* (Document ID 23408). https://www.cisco.com/c/en/us/support/docs/switches/catalyst-4000-series-switches/23408-140.html

Cisco. (2023, February 9). *Understand Rapid Spanning Tree Protocol (802.1w)* (Document ID 24062). https://www.cisco.com/c/en/us/support/docs/lan-switching/spanning-tree-protocol/24062-146.html

Droms, R. (1997). *Dynamic Host Configuration Protocol* (RFC 2131). RFC Editor. https://www.rfc-editor.org/info/rfc2131

Fette, I., & Melnikov, A. (2011). *The WebSocket protocol* (RFC 6455). RFC Editor. https://www.rfc-editor.org/info/rfc6455

FRRouting Project. (n.d.). *OSPFv2*. In *FRR user guide*. Retrieved October 11, 2026, from https://docs.frrouting.org/en/latest/ospfd.html

Fuller, V., & Li, T. (2006). *Classless Inter-domain Routing (CIDR): The Internet address assignment and aggregation plan* (RFC 4632). RFC Editor. https://www.rfc-editor.org/info/rfc4632

Jones, M., Bradley, J., & Sakimura, N. (2015). *JSON Web Token (JWT)* (RFC 7519). RFC Editor. https://www.rfc-editor.org/info/rfc7519

Moy, J. (1998). *OSPF version 2* (RFC 2328). RFC Editor. https://www.rfc-editor.org/info/rfc2328

OpenWrt Project. (n.d.-a). *DHCP and DNS configuration /etc/config/dhcp*. OpenWrt Wiki. Retrieved October 11, 2026, from https://openwrt.org/docs/guide-user/base-system/dhcp

OpenWrt Project. (n.d.-b). *Firewall configuration /etc/config/firewall*. OpenWrt Wiki. Retrieved October 11, 2026, from https://openwrt.org/docs/guide-user/firewall/firewall_configuration

OpenWrt Project. (n.d.-c). *Switch configuration*. OpenWrt Wiki. Retrieved October 11, 2026, from https://openwrt.org/docs/guide-user/network/vlan/switch_configuration

Ramírez, S. (n.d.). *FastAPI*. Retrieved October 11, 2026, from https://fastapi.tiangolo.com/

Rekhter, Y., Moskowitz, B., Karrenberg, D., de Groot, G. J., & Lear, E. (1996). *Address allocation for private internets* (RFC 1918). RFC Editor. https://www.rfc-editor.org/info/rfc1918

Srisuresh, P., & Egevang, K. (2001). *Traditional IP network address translator (Traditional NAT)* (RFC 3022). RFC Editor. https://www.rfc-editor.org/info/rfc3022

The Go Authors. (n.d.). *Documentation*. The Go Programming Language. Retrieved October 11, 2026, from https://go.dev/doc/